

The SAP S/4HANA Migration Playbook

A decision-grade guide for CIOs, CFOs, and transformation leaders planning the move off SAP ECC ahead of the 2027 end-of-mainstream-maintenance deadline. It covers approach selection, deployment models, the SAP Activate roadmap, readiness, cost & ROI, risk, and change management.

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HOW TO READ THIS PLAYBOOK

Statements drawn from external research are attributed inline and numbered to the source list in the Appendix (e.g. ^[5]). Where SAP's own guidance and independent analyst or user-group views diverge, that divergence is flagged in a teal "Where sources differ" box. Synthesis and recommendations that are the authors' own are presented without a source marker. Figures are planning benchmarks, not quotes for any specific organization.

1 Executive Summary

SAP will end mainstream maintenance for the core of SAP Business Suite 7, including SAP ERP 6.0 (ECC) Enhancement Packages 6–8, on **31 December 2027**, with an optional, paid extended-maintenance bridge running only to **31 December 2030**.^{[1][2]} For most organizations still on ECC, this is the defining IT decision of the decade, and the window to execute it well is closing.

2027

End of mainstream maintenance for ECC 6.0 EHP 6–8^[1]

~17,000

ECC customers Gartner projects will still be unmigrated by 2027^[5]

60%+

Of migration projects exceed budget, schedule, or both^[24]

12–24

Months typical for a single full S/4HANA program^[7]

What is at stake

S/4HANA is not a version upgrade; it is a re-platforming onto SAP's in-memory HANA database with a simplified data model, the Fiori user experience, and embedded AI. After 2027 (or 2030 for those who buy extended maintenance), an ECC system that has not moved transitions automatically to **customer-specific maintenance**: the same fee, but no new legal or regulatory updates, no new security patches, and only minimal bug fixes.^{[2][5]} The exposure is therefore not a sudden outage but a slow accumulation of compliance, security, and competitiveness risk, compounded by a tightening market for SAP skills.

The decisions leadership must own

1. **Approach:** Greenfield (new implementation), Brownfield (system conversion), or Bluefield / Selective Data Transition (hybrid). This sets the program's cost, risk, timeline, and how much legacy complexity you carry forward (Section 3).
2. **Deployment model:** on-premise, private cloud via *RISE with SAP*, or public cloud via *GROW with SAP*. This shapes total cost of ownership, control, and your innovation cadence (Section 4).
3. **Timing:** how to sequence the program backward from 2027, and what to do if you are starting late (Section 12).

Key takeaways

- **This is a business transformation, not an IT project.** Industry data consistently shows that organizations treating migration as a technical upgrade are the ones that overrun: roughly 30% of projects run about a third longer than planned and around 65% report serious post-go-live quality issues.^{[24][28]}
- **Readiness-first beats date-first.** Commit budget and go-live dates *after* an evidence-based readiness assessment (SAP Readiness Check, custom-code and data analysis), not before. The single largest source of overrun is data and custom-code work discovered late.^{[26][28]}
- **The skills market is the hidden clock.** The SAP consulting market exceeded US\$16 billion in 2025; specialist S/4HANA day rates have risen 30–50% versus 2022 as demand outstrips supply, and that pressure intensifies the closer organizations crowd toward 2027.^[26] Starting earlier is cheaper, not just safer.

- **Pure greenfield is the minority path.** Across independent surveys, brownfield and hybrid/selective approaches dominate; full greenfield is typically chosen by a minority of organizations, though surveys disagree on the exact split (Section 3).^{[9][25][15]}
- **Treat "RISE" economics with a buyer's eye.** SAP positions RISE with SAP as the strategic destination; independent advisors urge customers to unbundle the offer, negotiate price-escalation caps, and confirm it fits before committing (Section 4).^[27]

BOTTOM LINE FOR THE BOARD

If your organization has not yet completed an S/4HANA readiness assessment and locked an approach decision, that is the action to authorize this quarter. Even a brownfield conversion typically needs 6–18 months, and a greenfield re-implementation 18–36 months; working backward from a 2027 cutover, the runway for a comfortable, well-tested program is already short.^{[14][28]}

2 The Deadline & Maintenance Landscape

The first job of any program is to be precise about *which* deadline applies, because several dates circulate and the confusion has real planning consequences. The dates below reflect SAP's published maintenance strategy as summarized by SAP, ASUG, and independent commentators.

2.1 The dates that matter

Release / phase	Key date	What it means
ECC 6.0, EHP 0–5	Ended 31 Dec 2025	Mainstream maintenance has already ended . These systems are now in customer-specific maintenance and have no extended-maintenance option. ^{[2][5]}
ECC 6.0, EHP 6–8	31 Dec 2027	End of mainstream maintenance, the deadline that applies to the majority of remaining ECC customers. ^{[1][2]}
Extended maintenance (EHP 6–8)	1 Jan 2028 – 31 Dec 2030	Optional, separately contracted, at roughly +2 percentage points on the maintenance base (about a 9% increase in the annual maintenance fee). Scope is similar to mainstream, with some technical limits. ^{[2][3]}
Customer-specific maintenance	Automatic afterward	Same fee, but no new legal/regulatory updates, no new security patches , minimal bug fixes. Compliance and security exposure grows over time. ^{[2][5]}
S/4HANA (2023 release and later)	Maintained to at least 2040	From the 2023 release, mainstream maintenance was extended to 7 years (from 5), releases move to a two-year cadence , and SAP has committed to maintaining at least one release until 31 Dec 2040. ^[1]

OFTEN-MISSED: COMPATIBILITY PACKS

Independent of the maintenance dates, the limited-use rights that let classic ECC functionality run inside an S/4HANA on-premise system (the "Compatibility Packs") expire for most customers, with SAP having shifted the general transition deadline to **31 May 2026** and a small set of areas (e.g., Transportation Management, certain PP-PI functions) extended to end-2030.^{[6][7]} The practical effect: today's conversions must address functional adjustments and the technical move *together*, where earlier projects could defer them. Build this into scope.

2.2 Extended bridges, and their strings

For a narrow set of large, complex customers, SAP announced in the first half of 2025 a cloud-subscription bridge, the *SAP ERP, private edition, transition option*. It can extend ECC support beyond 2030, but only for customers who sign a RISE with SAP agreement, move to the HANA database before the end of 2030, and adjust third-party components (such as Java) that lose support.^{[4][3]} SAP has been explicit that this is *not* a general prolongation of the 2027/2030 dates.^[4]

WHERE SOURCES DIFFER: "IS MIGRATION MANDATORY?"

SAP's framing, reinforced by partners, treats the move to S/4HANA as the default next step. Independent voices push back: third-party support providers (e.g., Rimini Street) offer to maintain ECC well beyond SAP's dates, typically at a substantial discount, which keeps systems running but forgoes SAP innovation and SAP-delivered compliance updates. [3] User groups (DSAG, ASUG) accept the destination but resist the timeline pressure, warning that the deadline must trigger transformation rather than panic. [8] Speculation about a further extension to 2033 has circulated in the trade press, but SAP has publicly held the 2027/2030 line; treat any further extension as unconfirmed. [8]

2.3 The risk of inaction

- **Compliance and security drift.** Once in customer-specific maintenance, you stop receiving the legal, tax, and regulatory updates that keep payroll, statutory reporting, and e-invoicing current, a particular problem for regulated and multi-jurisdiction operations. [5][7]
- **A resource crunch you do not control.** Gartner estimated that only ~39% of SAP's roughly 35,000 ECC customers had even licensed an S/4HANA transition by end-2024, and projects on the order of 17,000 organizations still unmigrated at the 2027 deadline and 13,000+ still on ECC in 2030. [5] As that backlog converges on the same consultants, day rates rise and lead times lengthen. [7][26]
- **Innovation opportunity cost.** Capabilities such as real-time analytics on the Universal Journal, the Fiori experience, and SAP's Joule AI assistant are built for S/4HANA and are not available on the aging ECC stack. [40]
- **Compounding technical debt.** Every year of deferral adds custom code and data that will eventually need remediation anyway, raising, not lowering, the ultimate cost of the move.

3

Migration Approach Decision Framework

There are three principal routes to S/4HANA. SAP's own transition scenarios name them **New Implementation**, **System Conversion**, and **Selective Data Transition**; the market more often uses **Greenfield**, **Brownfield**, and **Bluefield**.^{[30][11]} ("Bluefield" is a trademark of SNP; SAP's neutral term is Selective Data Transition, or SDT.^{[9][20]})

3.1 The three paths at a glance

	Greenfield (New Implementation)	Brownfield (System Conversion)	Bluefield / SDT (Hybrid)
Core idea	Build a fresh S/4HANA system; re-design processes to SAP best practice; migrate only selected master data and open items. ^[20]	Convert the existing ECC system in place; retain configuration, history, and custom code. ^{[9][11]}	Stand up a new system, then selectively migrate the data, config, and objects worth keeping. ^{[16][11]}
Carries history?	No, by default (archive / access via ILM). ^[20]	Yes, all of it. ^[9]	Selective: for example, 3 years of transactions, all config. ^[16]
Typical duration	12–18 months (single country); 18–36 for large/global. ^{[9][28]}	6–18 months. ^{[9][14]}	9–18 months. ^[9]
Best for	Process redesign, M&A carve-outs, escaping heavy legacy debt, non-SAP migrations. ^[15]	Stable processes, limited change capacity, tight deadline, single ECC instance. ^[15]	System consolidation, carve-outs/divestitures, partial modernization, large data-volume reduction. ^[16]
Clean-core fit	Strongest: aligns naturally with clean core. ^[10]	Weakest: risks carrying legacy customizations and technical debt forward. ^{[9][10]}	Strong: a chance to offload custom code to BTP extensions selectively. ^[10]

3.2 Decision matrix

The matrix below scores each path on the dimensions leadership weighs. Ratings are directional; the right answer depends on your data condition, custom-code volume, change appetite, and deadline.

Dimension	Greenfield	Brownfield	Bluefield / SDT
Business / transformation value	High	Low	Med-High
Upfront cost	Highest	Lowest	Medium
Long-term TCO / debt	Lowest	Higher (debt)	Medium
Speed to go-live	Slowest	Fastest	Medium
Execution / delivery risk	Med-High	Medium	Medium
Change-management burden	Highest	Lowest	Medium
Data-migration complexity	Lower (selective)	Bundled in conversion	Highest (referential integrity)
Tooling / partner dependency	SAP best-practice content	SUM / DMO	SNP & specialist tools

3.3 What organizations are actually choosing

Adoption data is useful but inconsistent, a point worth understanding before quoting any single figure.

WHERE SOURCES DIFFER: THE APPROACH MIX

ASUG's 2024 member research put the split at roughly **34% brownfield / 33% greenfield / 20% hybrid**.^[25] An ISG 2026 dataset reported **34% brownfield / 18% greenfield / ~48% bluefield**, with selective transition the plurality.^[9] A LeanIX–PwC study found **44% brownfield / 14% greenfield / 42% hybrid**.^[15] The numbers disagree on the greenfield-versus-hybrid share, partly because "bluefield/hybrid" spans very different things, from a simple shell conversion to a multinational mix-and-match.^[9] The robust conclusion across all three: **pure greenfield is a minority choice; brownfield and selective/hybrid approaches carry most of the market.**

3.4 A practical way to choose

- 1. Start from change capacity, not technology.** If the organization is already absorbing major change (operating-model redesign, a global template, M&A), greenfield can ride that wave. If stability is paramount and change capacity is thin, brownfield is the realistic fit.^[13]
- 2. Let the data and custom-code inventory decide.** High Z-object volume and dense history pull toward brownfield by default, but that is also the path most exposed to a "clean-core deficit." A large, messy estate where only parts are worth keeping is the classic Bluefield case.^{[9][16]}
- 3. Map landscape events.** Carve-outs, divestitures, and multi-ERP consolidations are textbook Selective Data Transition scenarios.^[16]
- 4. Check the deployment constraint.** If the destination is public cloud (GROW), greenfield is effectively your only route (Section 4).^[36]

THE HONEST CAVEAT ON APPROACH

Recovery specialists who are brought into failing programs report that the greenfield-versus-brownfield choice is rarely the actual cause of failure; **execution and end-to-end ownership are.**^[33] Choose deliberately, then invest disproportionately in governance, testing, and a single accountable transformation leader.

4 Deployment Model Decision

Separate from *how* you migrate is *where* S/4HANA will run. In 2025 SAP grouped its cloud editions under a single "SAP Cloud ERP" umbrella, but the editions remain distinct in delivery and audience.^[20] The commercial packages are **RISE with SAP** (private cloud) and **GROW with SAP** (public cloud); classic **on-premise / BYOL** remains available but is being de-emphasized.^[22]

4.1 The three models compared

	On-premise / BYOL	RISE with SAP (Private Cloud)	GROW with SAP (Public Cloud)
What it is	You license and run S/4HANA on infrastructure you manage (own data centre or a hyperscaler).	S/4HANA Cloud, Private Edition as a bundled subscription: software, infrastructure, managed services, BTP credits, Signavio, Business Network. ^[22]	S/4HANA Cloud, Public Edition: multi-tenant SaaS, fully operated by SAP. ^{[31][35]}
Customization	Full control.	Deep customization retained; dedicated tenant; custom code possible. ^[37]	Fit-to-standard; extensions only via BTP / in-app extensibility. ^[39]
Migration paths	Greenfield, brownfield, SDT.	Brownfield <i>and</i> greenfield (and SDT). ^[36]	Greenfield only. ^{[36][35]}
Upgrade cadence	You choose.	Typically annual. ^[21]	Two mandatory releases per year (Feb & Aug). ^[21]
Innovation access	Slowest.	Fast.	First: SAP ships innovations (e.g., Joule) to public cloud first. ^{[40][39]}
TCO profile	Highest internal run cost.	Predictable subscription; higher than public for equivalent scope.	Lowest TCO; standardized. ^[34]
Best fit	Strict sovereignty or unusual control needs; transitional.	Large/complex enterprises migrating from ECC with significant custom development; regulated industries. ^{[37][34]}	Mid-market and net-new SAP adopters comfortable with standard processes. ^{[33][37]}

4.2 How to read the trade-off

The decision reduces to a familiar tension: **standardization and lower TCO** (public cloud / GROW) versus **control and customization** (private cloud / RISE).^[32] Public cloud rewards organizations willing to adopt SAP's best-practice processes and accept a fixed, twice-yearly release rhythm; it is the natural landing spot for first-time SAP adopters and simpler operations.^[37] Private cloud exists precisely for the enterprises whose proprietary processes, regulatory posture, and custom code make a shared, standardized environment impractical, which describes most large ECC incumbents.^[37] A practical qualifier: SAP requires a Digital Discovery Assessment (DDA) to confirm that

the scope items an organization needs are actually available in the public-cloud edition before committing to GROW.^[39]

WHERE SOURCES DIFFER: RISE ECONOMICS AND LOCK-IN

SAP presents RISE with SAP as the strategic destination, and many incumbents adopt it. Independent advisors are more cautious. Some organizations conclude RISE does not fit their cost or flexibility requirements.^[24] Buyer-side specialists recommend treating RISE as a negotiation, not a price list: **unbundle the offer** (which modules, infrastructure tier, BTP credits, and Signavio entitlement are included), and **negotiate a cap on the annual price-escalation clause at signature**, since the difference can run to seven figures over a five-year term and the clause is far harder to change at renewal.^[27] The diverging roadmaps of public and private editions (public increasingly gets new features first) is itself a strategic input.^[39]

RECOMMENDATION PATTERN

For most large ECC incumbents, the realistic pairing is **brownfield or selective transition onto RISE / private cloud**, optionally as a staged path that defers deeper process redesign. For mid-market and net-new adopters, **greenfield onto GROW / public cloud** typically delivers the best TCO and fastest innovation. The on-premise route is best reserved for genuine sovereignty constraints and is increasingly a transitional choice.

5

Phased Roadmap with SAP Activate

SAP Activate is SAP's implementation framework and the backbone of the RISE/GROW delivery methodology. It rests on three pillars (SAP Best Practices, which are ready-to-run reference processes; Guided Configuration; and an agile delivery model) and organizes work into six phases, each closed by a quality gate.^{[13][14]} The phases are **Discover** → **Prepare** → **Explore** → **Realize** → **Deploy** → **Run**.^[26]

5.1 The six phases

Phase	Objective	Key activities & outputs	Typical duration*
Discover	Understand S/4HANA's value; define target architecture and the business case.	Solution trials; as-is process & landscape mapping; SAP Readiness Check; deployment-model and approach decision. <i>Output:</i> value case, high-level roadmap. ^{[15][14]}	4–8 weeks
Prepare	Mobilize the project; stand up environments and governance.	Project charter, scope, RACI, plans; sandbox/starter systems; kickoff; team onboarding. <i>Output:</i> baselined plan, governance, environments. ^{[14][28]}	4–8 weeks
Explore	Confirm the solution via fit-to-standard; capture gaps.	Fit-to-standard workshops on Best Practices; delta-config and gap capture (WRICEF/RICEFW); design docs (KDS/BPD); master-data & test planning. <i>Output:</i> backlog, signed-off design, Q-gate. ^{[23][30]}	2–4 months
Realize	Build, configure, and test iteratively in agile sprints.	Configuration; WRICEF development; data migration build; unit, string, integration, and user-acceptance testing; cutover plan drafting. <i>Output:</i> validated, release-ready solution. ^{[14][30]}	4–9 months
Deploy	Cut over to production and transition to operations.	Final readiness checks; data load; role/authorization validation; go-live cutover; hypercare. <i>Output:</i> live production system. ^{[23][30]}	2–6 weeks (+ hypercare)
Run	Stabilize, support, and continuously improve.	Incident resolution; performance tuning; adoption tracking; uptake of new releases and innovations. <i>Output:</i> steady-state operations, benefit realization. ^{[25][28]}	Ongoing

**Durations are indicative for a single full S/4HANA program; brownfield conversions compress Explore/Realize, while global greenfield rollouts extend them. SAP Activate is delivered through SAP Cloud ALM (cloud projects) or legacy Solution Manager, with the Activate Roadmap Viewer providing phase-by-phase, role-based guidance and quality gates.^{[26][25]}*

5.2 RACI by phase (illustrative)

A working RACI keeps accountability explicit. R = Responsible, A = Accountable, C = Consulted, I = Informed. Adapt to your governance.

Activity / phase	Exec Sponsor / Steering	Program / Transf. Lead	Business Process Owners	IT / Basis & Security	SI / Implementation Partner
Business case & approach (Discover)	A	R	C	C	C
Governance & environments (Prepare)	A	R	I	R	C
Fit-to-standard & design (Explore)	I	A	R	C	R
Build, config & test (Realize)	I	A	C	R	R
Cutover & go-live (Deploy)	A	A	C	R	R
Hypercare & optimize (Run)	I	A	C	R	C

DISCIPLINE THAT PROTECTS THE TIMELINE

The fit-to-standard workshops in **Explore** are where scope is won or lost. The proven discipline is to ask, for every requested deviation, *"why won't the standard best practice work for us?"* Each accepted gap becomes WRICEF/RICEFW work that must be built, tested, and maintained. In a clean-core world it should live as a side-by-side extension on BTP rather than a core modification. ^{[10][31]}

Readiness Assessment & Pre-Migration Activities

The readiness phase is where overruns are prevented. The mantra is **readiness-first**: produce an evidence base before committing budget and dates, because the costliest surprises, data quality and custom code, are the ones discovered late.^{[26][28]}

6.1 Technical prerequisites you must confirm

- **HANA only.** S/4HANA runs exclusively on the SAP HANA in-memory database. If your source is on Oracle, DB2, or SQL Server, the database migration happens during conversion via the *Database Migration Option (DMO)* of the Software Update Manager (SUM).^[16]
- **Unicode only.** S/4HANA is Unicode-only. A non-Unicode system must be converted to Unicode *first*; an integrated non-Unicode-to-Unicode conversion is not available for current targets (the Combined Upgrade & Unicode Conversion, CU&UC, handles this as a separate step).^{[16][18]}
- **Source release.** A single-step conversion is possible only if you are on ECC 6.0, on an ABAP-only (not dual ABAP/Java) stack, and already Unicode. Older or non-Unicode systems require a two-step path: upgrade first, then convert.^{[18][35]} The EHP level itself does not block conversion.^[18]
- **Add-ons & business functions.** All active add-ons and business functions must be compatible with the target; the Maintenance Planner verifies this and generates the stack file SUM consumes.^{[17][68]}
- **Business Partner conversion.** S/4HANA mandates the Business Partner as the leading object; converting customers and vendors (Customer/Vendor Integration, CVI) is a required preparatory step.^{[35][66]}

6.2 The readiness toolset

Tool	Purpose
SAP Readiness Check	The headline analysis: from a metadata extract of your system, it surfaces relevant simplification items, custom-code findings, add-on compatibility, sizing, recommended Fiori apps, and more. ^{[16][30]}
Simplification Item Check (SI-Check)	Run in the source system and called by SUM; identifies the simplification items relevant to your conversion and flags inconsistencies that would break the SUM run. ^[17]
Maintenance Planner	Validates add-on/business-function compatibility and produces the stack XML required by SUM. ^{[17][68]}
Custom Code Migration app / ATC / SCMON	Inventories custom (Z) objects, measures actual usage, and identifies what must be remediated, retired, or refactored for S/4HANA. ^{[16][9]}
SUM with DMO	The technical conversion engine: upgrades the application to S/4HANA and migrates the database to HANA in one orchestrated run; downtime-optimized and near-zero-downtime variants reduce the cutover window. ^{[16][64]}
SAP S/4HANA Migration Cockpit / SLT	Native data-load tooling (greenfield loads) and real-time replication, respectively; for complex selective transitions, specialist platforms such as SNP are common. ^{[16][11]}

6.3 Custom code & the clean core

The average enterprise carries on the order of 2,000+ custom ABAP objects, and unmanaged custom code is a leading cause of delay.^[59] The clean-core principle is SAP's architectural answer: keep the digital core close to standard and move custom functionality to side-by-side extensions on SAP BTP (or governed key-user/developer extensibility) rather than modifying the core.^{[9][10]} Practically: use usage data to **decommission dead code**, remediate what is genuinely needed, and treat the migration as the once-in-a-decade opportunity to shed accumulated debt, an opportunity brownfield in particular must consciously seize or forfeit.^{[66][9]}

6.4 Data cleansing, archiving & UX

- **Data quality.** Cleanse master data (the duplicate-vendor problem is a classic and expensive failure mode), validate referential integrity, and reconcile before load. Budget roughly 10–15% of total project cost for data work; it is where budgets quietly balloon.^{[26][54]}
- **Archiving & footprint reduction.** Archive and decommission ahead of migration to shrink the data set you carry. One documented program cut an Oracle database by over 80% (2.9 TB) before moving, using SAP Information Lifecycle Management.^[32] Less data means faster, cheaper, lower-risk migration.
- **Fiori / UX & security.** Plan the Fiori launchpad and role-based access early; S/4HANA's role-based UX requires re-mapping legacy authorizations to new objects rather than lifting roles across unchanged.^[11]

7 Tools, Templates & Checklists

The instruments below are ready to adapt. They are deliberately lightweight: their value is in forcing the right conversations at the right gate.

7.1 Readiness scorecard

Score each dimension 1–5 (1 = not ready, 5 = fully ready). An average below ~3.0 signals that readiness work, not a go-live date, is the immediate priority.

Dimension	What "ready" (5) looks like	Score
Executive sponsorship	Named accountable sponsor; funded; value targets agreed at board level.	___ / 5
Business case & benefits	Quantified benefits, costs, and contingency; benefit owners assigned.	___ / 5
Data quality & volume	Profiled, cleansed plan in place; archiving strategy defined.	___ / 5
Custom code	Inventoried with usage data; remediation/retirement plan set.	___ / 5
Technical prerequisites	Unicode, source release, add-ons, Business Partner confirmed.	___ / 5
Approach & deployment	Greenfield/brownfield/SDT and RISE/GROW/on-prem decided with rationale.	___ / 5
Skills & partner	SI selected; internal capability and backfill planned.	___ / 5
Change & adoption	Impact assessment, training, and comms plan resourced.	___ / 5

7.2 Business case / ROI template

Line	Notes
One-time program cost	SI/consulting (40–60% of total), data work (~10–15%), licences, infrastructure/dual-run, training, contingency. ^{[26][51]}
Recurring cost change	RISE/GROW subscription or maintenance vs. current ECC run cost; model years 1–5 with any escalation cap. ^[27]
Avoided cost	Retired legacy licences/hardware; reduced maintenance after decommissioning. ^{[29][32]}
Productivity benefit	Process cycle-time gains, faster close, end-user time saved. ^[29]
Revenue / risk benefit	New capabilities, real-time decisioning, compliance risk avoided. ^[29]
Net value & payback	NPV, IRR, payback period; sensitivity on the largest assumptions.

7.3 Stakeholder map (archetypes)

Stakeholder	Primary interest	Engagement
CFO	Cost certainty, controls, statutory reporting, faster close.	Co-owns business case; steering.
CIO / CTO	Architecture, run cost, risk, talent.	Accountable for delivery platform.
COO / business unit leads	Process continuity, adoption, disruption.	Process-owner network; UAT.
CISO	Security posture, authorizations, compliance.	Consulted on roles & controls.
End users	Usability, training, day-one productivity.	Champions, training, hypercare.
SI / partner	Scope, delivery, knowledge transfer.	Contracted; governed by Q-gates.

7.4 Go / no-go checklist (pre-cutover)

- All integration and user-acceptance test cycles passed; defect leakage within threshold.
- At least one full mock cutover (ideally more) completed within the downtime window, dress-rehearsed.
- Data migration validated and reconciled; fallback/rollback plan tested.
- Roles and authorizations validated against new Fiori-based access model.
- Cutover runbook with owners, timings, and decision points; command-centre staffed.
- Hypercare team, severity definitions, and escalation paths in place.
- Business sign-off from each process-owner area; sponsor go-decision recorded.

7.5 Cutover plan template (skeleton)

Window	Activities	Owner
T-14 to T-1 days	Freeze transports; final pre-checks; comms; transaction freeze plan; data extract.	Program / Basis
T-0 (downtime)	Business freeze; SUM/DMO conversion or data load; conversions; technical validation.	Basis / SI
T-0 (validation)	Reconciliation; smoke tests; authorization checks; key business validations.	Process owners
Go-live	Sponsor go-decision; open system; first transactions monitored.	Sponsor / Program
Hypercare (1-6 wks)	Command centre; rapid defect triage; performance tuning; adoption support.	Program / SI

Downtime-optimized DMO and near-zero-downtime techniques can compress the technical cutover; some programs report cutovers of hours rather than the multi-day windows of classic conversions.^{[64][59]}

8 KPIs, Costs & ROI

8.1 Cost benchmarks

Costs vary enormously with size, data condition, custom-code volume, and approach, which is exactly why a readiness-based estimate beats a benchmark. As planning anchors:

Segment / approach	Indicative range	Timeline
Mid-market (total program)	~US\$1M–\$8M, depending on complexity and path. ^[26] Some estimates put mid-market consulting alone at \$0.8M–\$3M. ^[27]	9–18 months
Large enterprise	Routinely >US\$10M; \$50M+ for the largest programs; Fortune-500 transformations have reached ~\$400M over multiple years. ^{[27][30]}	18–36+ months
Brownfield (conversion)	Lowest: low six to low seven figures. ^[28]	9–18 months
Greenfield (re-implementation)	Higher: seven to eight figures. ^[28]	18–36 months

Cost composition is consistent: **implementation consulting is the largest line at 40–60% of total**, followed by data work (~10–15%), licences/subscription, infrastructure and dual-run, training, and contingency.^{[26][27]}

THE RATE-INFLATION CLOCK

The SAP consulting market passed US\$16 billion in 2025 and is projected to approach \$39 billion by 2035; specialist S/4HANA day rates are up 30–50% versus 2022 as demand runs at roughly three times supply, pressure that intensifies as more customers crowd toward 2027.^[26] Every quarter of delay tends to raise the rate you eventually pay.

8.2 What "success" looks like: KPI set

Category	Representative KPIs
Delivery	On-time / on-budget variance; defect leakage to production; cutover downtime (hours); mock-cutover pass rate.
Adoption	Active-user rate; training completion; help-desk ticket volume and trend post-go-live.
Process value	Financial close days; order-to-cash / procure-to-pay cycle times; on-time reporting.
Architecture health	Custom-code reduction %; share of extensions on BTP vs. core mods (clean-core ratio); data footprint reduction.
Benefit realization	% of business-case benefits tracked and realized; TCO change vs. baseline.

8.3 Value drivers & ROI evidence

The durable value of S/4HANA comes from the simplified data model (e.g., the Universal Journal), real-time analytics, the Fiori experience, lower run-time TCO, embedded AI, and, increasingly, readiness for AI-driven processes that the ECC stack cannot support.^{[40][9]} A vendor-commissioned Forrester Total Economic Impact analysis modelled, for a composite adopter, a three-year revenue uplift of roughly \$156M (+5.8%), reduced customer churn worth ~\$8.4M, end-user productivity gains of ~\$2.1M, and avoided software/hardware cost.^[29]

WHERE TO APPLY SKEPTICISM

Treat vendor-commissioned ROI models (including Forrester TEI studies) as **illustrative of value categories, not as guaranteed outcomes**; the composite figures reflect favorable adopters. Set against them the hard counter-evidence: independent surveys find ~49% of live adopters exceeded their original budget (up 17 points year-on-year), with consulting fees the main driver.^[25] A credible business case carries both the upside and a realistic contingency.

9 Risks, Pitfalls & Mitigations

The failure modes are well documented and remarkably consistent: industry research finds 60%+ of programs exceed budget, schedule, or both; roughly 30% run about a third longer than planned; and around 65% report severe-to-very-severe quality deficiencies after go-live.^{[24][28]} Almost all trace back to a handful of avoidable causes.

Pitfall	How it shows up	Mitigation
Scope creep / gold-plating	Every business unit re-requests its legacy customization; the backlog balloons.	Enforce fit-to-standard discipline at Explore; require a written justification for each deviation; route extensions to BTP, not the core. ^[10]
Data quality	Duplicate or invalid master data breaks processes, for example duplicate vendors causing delayed payments and supplier penalties. ^[54]	Profile and cleanse early; reconciliation and governance workflows; budget 10–15% for data; archive to shrink scope. ^{[26][32]}
Custom-code debt	Brownfield carries Z-objects forward, undermining clean core and future AI readiness. ^[9]	Inventory with usage data; decommission the unused; remediate the essential; move logic to side-by-side extensions. ^{[10][66]}
Under-testing	Quality issues surface in production; ~65% report serious post-go-live defects. ^[24]	Full integration + UAT cycles; multiple mock cutovers; automated regression where possible; defect-leakage thresholds in the go/no-go.
Change resistance	Low adoption; shadow processes; benefits never realized.	Early impact assessment; role-based training; visible quick wins; sponsor-led communications (Section 10). ^[33]
Skills shortage / rate inflation	Scarce, expensive specialists as 2027 nears; lead times stretch. ^{[7][26]}	Secure partner and key people early; build internal capability and knowledge transfer; start before the crowd.
"IT-only" framing	Treated as a technical upgrade; business disengages; value missed.	Run it as a business transformation with co-owning business sponsors and benefit owners. ^{[26][54]}
Date-before-data commitment	Budget and go-live fixed before measuring the estate; overruns follow.	Readiness-first: assess, then commit, with a real contingency for the ~30% of projects that slip. ^[28]

10 Change Management & Organizational Readiness

The recurring lesson from recovered programs is blunt: **the migration path matters less than the quality of leadership and execution.** Where programs were slipping, the decisive intervention was a hands-on, accountable leader aligning board and operational teams, tracking real adoption, and rehearsing go-live under pressure, not a change of technical approach.^[33]

10.1 Governance

- **Single accountable transformation leader** who owns the end-to-end arc, not the vendor, and not diffused across IT.^[33]
- **Executive steering** with CFO and CIO co-ownership and a standing cadence tied to the SAP Activate quality gates.
- **A business-process-owner network** empowered to make fit-to-standard decisions and sign off design.
- **A value-realization function** that tracks benefits from business case through the Run phase, so value is measured, not assumed.

10.2 Adoption & training

- **Role-based, hands-on training** on the new Fiori experience and changed processes, timed close to go-live to maximize retention.^[76]
- **Quick wins and real-world proof**, such as faster approvals and better insight, to convert skeptics, paired with sponsor-led communication.^[33]
- **A champions network** embedded in the business to support peers through hypercare.
- **Tap the community:** user groups (ASUG, DSAG) and SAP enablement provide playbooks, peer benchmarks, and candid lessons.^{[8][25]}

OPERATING PRINCIPLE

Adopt explicit guiding principles and hold the line on them. One divested business ran its greenfield program on four words (standardize, harmonize, simplify, scalable) and used them to challenge every request to deviate, standing up its independent S/4HANA platform in 15 months.^[31] Principles like these turn thousands of micro-decisions into one consistent answer.

Case Examples

The examples below are representative migration stories drawn from practitioner accounts and published partner case studies; some are anonymized in their original sources, and partner-reported outcomes should be read as vendor-reported. They are included for their *lessons*, not as endorsements.

Large diversified group: brownfield, rehearsed cutover

A steadily acquisitive group of 600+ entities and ~75,000 employees ran a technical (brownfield) conversion from SAP Business Suite on HANA to S/4HANA, a roughly 10-month effort culminating in a three-day cutover weekend.

^[31] **Lesson:** for large, stable estates, a technical-first conversion with an exhaustively rehearsed cutover de-risks the move; optimization can follow once you are on the new platform.

Divested chemicals business: greenfield as a clean slate

A gas-and-chemical business spun off from a much larger parent abandoned the parent's heavily customized ECC environment and ran a big-bang greenfield implementation, standing up its own infrastructure in ~15 months under the principles standardize/harmonize/simplify/scalable.^[31] **Lesson:** carve-outs and divestitures are a natural greenfield trigger, a rare chance to start clean without legacy constraints.

Aerospace manufacturer: Bluefield consolidation

Facing five separate local ERP systems, a major commercial-aircraft manufacturer used a selective (Bluefield) transition to centralize local finance processes onto a single aligned S/4HANA platform.^[12] **Lesson:** selective data transition is purpose-built for consolidating multiple ERPs while keeping only the configurations and data worth retaining.

Specialty-chemicals manufacturer: data rightsizing first

Before migrating, one manufacturer reduced its Oracle database by over 80% (a 2.9 TB reduction) through ILM archiving and legacy-content decommissioning.^[32] **Lesson:** shrinking the data footprint before migration cuts cost, downtime, and risk. Data strategy is a migration accelerator, not an afterthought.

National rail operator: decommissioning value

A national rail operator phased out legacy systems as part of its data-management strategy and reported a 50% reduction in associated maintenance costs.^[32] **Lesson:** retiring legacy systems is a quantifiable benefit line in the business case, not just hygiene.

Tier-1 automotive supplier: recovery through leadership

A European Tier-1 automotive supplier on a greenfield program saw adoption lag and go-live slip quarter after quarter; installing a hands-on program manager turned it around within four months.^[33] **Lesson:** the most powerful intervention in a troubled program is end-to-end ownership and execution discipline, not switching approach.

12 Implementation Timeline & the "Starting Late" Scenario

Because a single full program typically takes 12–24 months, and a brownfield conversion can be done in as little as 6–18, a project starting in **mid-2026** can still reach a comfortable go-live before the end of 2027, with extended maintenance available as a fallback bridge if needed.^{[7][9]} The discipline is to plan *backward* from the deadline and protect a testing-and-stabilization buffer.

12.1 Backward-planned sequence (target: go-live before end-2027)

Window	Focus
Now – Q3 2026	Discover & decide. Readiness Check, custom-code & data assessment, business case, approach and deployment decisions, partner selection.
Q3 2026 – Q4 2026	Prepare & Explore. Mobilize, environments, fit-to-standard workshops, design sign-off, backlog.
Q4 2026 – Q3 2027	Realize. Configure, build, migrate data, and test in sprints; draft and rehearse cutover.
Q3 2027 – Q4 2027	Deploy & stabilize. Final mock cutovers, go-live, and hypercare, with buffer ahead of 31 Dec 2027.
2028+	Run & optimize. Adopt the half-yearly/annual release cadence; realize deferred process improvements and AI capabilities.

A useful rule of thumb: those starting after mid-2025/2026 may find a global greenfield tight without phased go-lives, whereas a focused brownfield conversion remains feasible on this runway.^[9]

12.2 If you are starting late

If a comfortable pre-2027 go-live is no longer realistic, sequence the options deliberately rather than panicking:

1. **Buy the bridge.** Elect extended maintenance (EHP 6–8 only) to 2030 to protect compliance and security while you execute, budgeting the roughly +2-point fee, but treat it as a runway, not a destination.^{[2][5]}
2. **Minimize scope to move.** Favor a technical-first brownfield conversion to get onto S/4HANA quickly, explicitly deferring optimization to the Run phase. This is the fastest compliant path.^{[35][9]}
3. **Phase the go-lives.** Break a large rollout into modular or regional waves to fit the runway and reduce per-event risk.^[75]
4. **For very large/complex estates,** assess the *SAP ERP, private edition, transition option*, the RISE-based bridge that can extend support beyond 2030, with its conditions (RISE contract, HANA by 2030, third-party-component adjustments).^{[4][3]}
5. **Consider third-party support as a deliberate stopgap,** not a strategy, understanding that it sustains operations but forgoes SAP innovation and SAP-delivered regulatory updates.^{[3][7]}
6. **Lock in scarce resources now.** The later you start, the tighter and pricier the consultant market; securing your partner and key people early is itself a risk control.^{[7][26]}

THE ONE THING NOT TO DO

Do not "wait and see." User groups and analysts agree the breathing room to 2030 is not a reason to delay but the signal to start, precisely because the consultant crunch, not the software deadline, is the binding constraint after 2027.^{[8][5]}

A

Appendix

A.1 Glossary of SAP terms

ABAP

SAP's proprietary programming language; "Z-objects" are customer-built ABAP objects layered on standard SAP.

ATC (ABAP Test Cockpit)

Static-analysis tool used to check custom code for S/4HANA readiness.

Bluefield / Selective Data Transition (SDT)

Hybrid migration that selectively moves chosen data, config, and objects to a new S/4HANA system. "Bluefield" is an SNP trademark; SDT is SAP's neutral term.

Brownfield / System Conversion

In-place conversion of an existing ECC system to S/4HANA, retaining config, data, and custom code.

BTP (Business Technology Platform)

SAP's platform for side-by-side extensions, integration, analytics, and AI; the clean-core home for custom functionality.

Business Partner (CVI)

The mandatory leading master-data object in S/4HANA; Customer/Vendor Integration converts legacy customers and vendors to it.

Clean core

The principle of keeping the digital core close to standard, building extensions on BTP rather than modifying the core, to keep upgrades simple.

DMO (Database Migration Option)

An option of SUM that migrates a non-HANA database to HANA during the conversion.

ECC (ERP Central Component)

The core of SAP Business Suite 7; the legacy ERP most organizations are migrating from.

Fiori

SAP's role-based, modern user-experience framework for S/4HANA.

Greenfield / New Implementation

A fresh S/4HANA build with redesigned processes; only selected data is migrated.

GROW with SAP

The commercial package for S/4HANA Cloud, Public Edition (multi-tenant SaaS), for mid-market and net-new adopters.

HANA

SAP's in-memory database; the mandatory foundation for S/4HANA.

ILM (Information Lifecycle Management)

SAP tooling for archiving, retention, and decommissioning of data.

RISE with SAP

The commercial package for S/4HANA Cloud, Private Edition (dedicated) plus services, for complex enterprises.

SAP Activate

SAP's implementation methodology: Discover, Prepare, Explore, Realize, Deploy, Run, with quality gates.

SAP Cloud ALM

Application Lifecycle Management used to run SAP Activate cloud projects (with quality gates).

SUM (Software Update Manager)

The technical engine that performs the S/4HANA conversion (with DMO when migrating the database).

Simplification Item

A documented functional/technical change in S/4HANA that a conversion must address; surfaced by the SI-Check and Readiness Check.

Universal Journal (ACDOCA)

The single, unified line-item table underpinning S/4HANA Finance and real-time reporting.

WRICEF / RICEFW

The catalog of custom development objects (Workflows, Reports, Interfaces, Conversions, Enhancements, Forms).

A.2 Sources

External sources used in this playbook, numbered as cited inline. Vendor and partner materials are labelled where relevant; independent analyst and user-group sources are noted as such. Cross-checked across at least two sources where feasible; flagged divergences appear in "Where sources differ" boxes.

1. SAP, ASUG and SoaPeople summaries of SAP maintenance phases. See "End of Support for SAP ECC and S/4HANA" (SoaPeople, 2026) and SAP maintenance documentation. soapeople.com
2. ASUG, "Your SAP Investment Unleashed: End of Mainstream Maintenance" (2025). asug.com
3. Rimini Street, "No Extension to ECC Support: 2027 Deadline" (2025). Extended-maintenance pricing, third-party support. riministreet.com
4. The Register, "SAP extends support deadline for getting off legacy software" (Jan 2025). Private-edition transition option. theregister.com
5. SoftwareSeni, "What SAP ECC End of Support Actually Means" (2026). Gartner figures and Di Capua remarks. softwareseni.com
6. SEIDOR, "Understanding SAP ECC Deadlines: 2026 and 2027" (2025). Compatibility Packs. seidor.com
7. ibsolution, "End of maintenance for SAP ECC in 2027" (2026). DSAG investment data, timeline guidance. ibsolution.com
8. E3 Magazine, "Deadline Extension for ECC 6.0" (2025). DSAG views; 2033 speculation. e3mag.com
9. SoftwareSeni, "Brownfield, Greenfield or Bluefield, with the Data" (2026). ISG 2026 mix, SDT detail. softwareseni.com
10. Kellton, "SAP S/4HANA Migration: Greenfield vs Bluefield" (2025). kellton.com
11. Basis Admin, "Brownfield vs Greenfield: Decision Guide" (2026). SDT tooling. basisadmin.com
12. SNP Group, "Greenfield, brownfield, and Bluefield." Airbus case (vendor). snpgroup.com
13. SAP, "SAP Activate: Accelerate SAP Cloud ERP Deployment" (sap.com).
14. SAP Learning, "Describing the Methodology Structure" and "Planning Your Implementation with SAP Activate."
15. LeanIX, SAP Activate and Greenfield-vs-Brownfield wiki, incl. LeanIX–PwC study (44/14/42). leanix.net
16. SAP Help Portal, "Conversion to SAP S/4HANA with SUM." HANA-only, Unicode-only, DMO. help.sap.com
17. SAP, "Conversion Guide for SAP S/4HANA 2023" (Jan 2026). SI-Check, two-step path, SDMI. help.sap.com
18. EPI-USE Labs, "Converting to SAP S/4HANA: your technical options" (2025). One-step prerequisites, CU&UC. epiuselabs.com
19. Talan, "RISE with SAP or GROW with SAP?" Positioning. talan.com
20. Accely, "Difference Between SAP S/4HANA Public and Private Edition" (2026). SAP Cloud ERP rebrand. accely.com
21. Nagarro, "Navigating SAP's cloud transition" (2024). Update cadence (Feb/Aug; annual private). nagarro.com
22. Arvato Systems, "SAP Cloud ERP: RISE and GROW in Comparison" (2025). arvato-systems.com
23. SAPinsider, "SAP S/4HANA Migration 2025 Benchmark Report" (Feb 2025). Adoption, BTP. sapinsider.org
24. Synovia Digital, "SAP S/4HANA Migration Trends Heading into 2026" (2025). Overrun/quality stats (per SAPinsider). synoviadigital.com
25. ASUG, "The State of SAP S/4HANA Adoption" (2024 Journeys research). Approach mix, budget overruns. asug.com
26. Tachyon, "SAP S/4HANA Migration Costs in 2026" (2026). Cost ranges, consulting market, rate inflation, data %. tachyontech.com
27. Redress Compliance, "SAP S/4HANA Migration Cost Estimator" (2026). Consulting ranges, RISE escalation cap. redresscompliance.com
28. Thinklytics, "SAP S/4HANA Migration Costs 2026" (2025). Approach cost/timeline, 65/60/55 stats. thinklytics.com
29. Fingent, "Cost-Benefit Analysis of SAP S/4HANA Migration" (2024). Forrester TEI figures (vendor-commissioned). fingent.com

30. Mind2Matter, "SAP S/4HANA Migration Cost" (2025). Range incl. large-enterprise figures. mind2matter.co
31. SAPinsider, "SAP S/4HANA Migration Lessons Learned" (2026). Anonymized customer cases. sapinsider.org
32. Proceed Group, "S/4HANA migration best practices." Croda and Dutch Rail data-management cases (vendor). proceedgroup.com
33. CE Interim, "SAP S/4HANA Migration: Greenfield or Brownfield?" (2025). Recovery case, execution lessons. ceinterim.com
34. PwC, "Journey to S/4HANA." Client survey. pwc.com
35. SAP-PRESS blog, "Considering an SAP S/4HANA System Conversion" (2024). One-/two-step, Business Partner. blog.sap-press.com
36. Accenture, "Conversion to SAP S/4HANA" brochure. Smart Brownfield (hybrid). accenture.com
37. SAP Community, "Brownfield SAP S/4HANA implementation" (2025). Conversion steps, prerequisites. community.sap.com
38. SAP Community, "S/4HANA Greenfield Implementation: Activate Methodology" (2026). Phase deliverables. community.sap.com
39. SAP Help Portal, "Conversion to SAP S/4HANA 2022/2023." Downtime-optimized conversion. help.sap.com
40. Nagarro and general SAP product material. ECC stack age, Joule AI availability on S/4HANA. nagarro.com

A.3 Further reading

- **SAP official:** SAP Activate methodology pages and the Activate Roadmap Viewer; the SAP S/4HANA Conversion Guide on the SAP Help Portal; SAP Readiness Check documentation; RISE with SAP and GROW with SAP product pages.
- **Big 4 & SIs:** Deloitte, PwC, EY, and KPMG S/4HANA readiness and value-case materials; Accenture and Capgemini conversion methodologies; IBM and Infosys transformation guidance.
- **Independent analysts & user groups:** Gartner and Forrester S/4HANA research; ASUG "S/4HANA Journeys" research; DSAG Investment Report; SAPinsider benchmark reports.

A NOTE ON EVIDENCE QUALITY

Deadline and methodology facts in this playbook draw on SAP's own documentation and are corroborated by multiple independent summaries. Cost, timeline, and outcome figures are **benchmarks synthesized across vendor, SI, and analyst sources and vary widely**; they should inform a readiness-based estimate for your organization, not substitute for one. ROI figures from vendor-commissioned studies are illustrative. Where a single claim rested on a single or partisan source, notably RISE economics and any post-2030 extension, that limitation is flagged in the body.